COHOMOLOGY OVER AN AFFINE ETALE NEIGHBOURHOOD IS THE GLOBAL SECTIONS OF THE HIGHER DIRECT IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The higher direct images commute with etale localization. Properties of Spaces, label spaces-properties-lemma-pushforward-etale-base-change-modules (invoked at proof line 468). It reduces the statement for an arbitrary Y to the case where $Y = V$ is affine.
- The Leray spectral sequence of a morphism of algebraic spaces. Cohomology on Sites, label sites-cohomology-lemma-Leray (invoked at proof line 474). It gives the spectral sequence $E_2^{p,q} = H^p(Y, R^qf_*\mathcal{F})$ converging to $H^{p+q}(X, \mathcal{F})$.
- Quasi-coherence of the higher direct images along a quasi-compact quasi-separated morphism. Same chapter spaces-cohomology.tex, label lemma-higher-direct-image (invoked at proof line 475). It puts $R^qf_*\mathcal{F}$ inside the class of quasi-coherent modules on the affine base, whose higher cohomology vanishes.
- Vanishing of the higher cohomology of quasi-coherent modules on an affine scheme. Cohomology of Schemes, label coherent-lemma-quasi-coherent-affine-cohomology-zero (invoked at proof line 478). It kills the terms $E_2^{p,q}$ with $p > 0$, so the spectral sequence degenerates at E_2 and the claim follows.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Navigation pointer for \ref{coherent-lemma-quasi-coherent-affine-cohomology-zero} used inside the counted span (line 478); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0720: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 339–339. Navigation pointer for \ref{lemma-higher-direct-image} used inside the counted span (line 475); the argument that consumes it is retained in the span.

Source reference. Original lemma 0732: sources/stacks-chow/tags.txt, source lines 9046–9046. Navigation pointer for \ref{sites-cohomology-lemma-Leray} used inside the counted span (line 474); the argument that consumes it is retained in the span.

Source reference. Original lemma 03LX: sources/stacks-chow/tags.txt, source lines 4729–4729. Navigation pointer for \ref{spaces-properties-lemma-pushforward-etale-base-change-modules} used inside the counted span (line 469); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a quasi-separated and quasi-compact morphism of algebraic spaces over S. For any quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ and any affine object V of $Y_{étale}$ we have*

$$H^q(V \times_Y X, \mathcal{F}) = H^0(V, R^q f_* \mathcal{F})$$

for all $q \in \mathbf{Z}$.

Proof. Since formation of Rf_* commutes with étale localization (Properties of Spaces, Lemma 03LX) we may replace Y by V and assume $Y = V$ is affine. Consider the Leray spectral sequence $E_2^{p,q} = H^p(Y, R^q f_* \mathcal{F})$ converging to $H^{p+q}(X, \mathcal{F})$, see Cohomology on Sites, Lemma 0732. By Lemma 0720 we see that the sheaves $R^q f_* \mathcal{F}$ are quasi-coherent. By Cohomology of Schemes, Lemma 01XB we see that $E_2^{p,q} = 0$ when $p > 0$. Hence the spectral sequence degenerates at E_2 and we win. □